

MotionCanvas

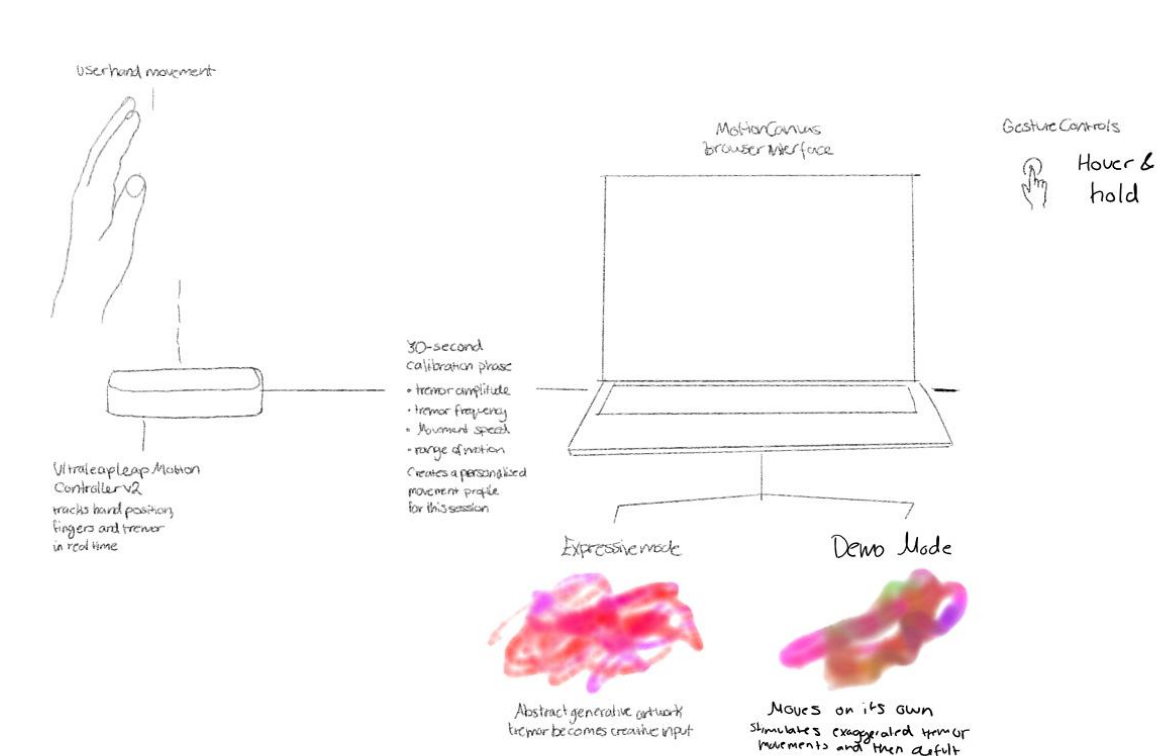
Group Document – Methods 4: Critical Interfaces for HCI

Github Repository: <https://git.arts.ac.uk/24016221/Critical-Interfaces-for-HCI.git>

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1. Introduction

MotionCanvas is a web-based digital drawing system designed for adults with motor differences, including essential tremor, multiple sclerosis, cerebral palsy, and limited range of motion. Existing creative software such as Adobe Fresco and Procreate, assumes precise, stable hand control and treats involuntary movement as error to be corrected. MotionCanvas challenges this assumption: movement is not a problem to suppress but the creative material itself. The interface captures natural hand movement via an Ultraleap Motion Controller and translates it into expressive digital artwork through a toggleable tremor reframe system. Three iterative versions were developed, building from a basic HTML5 proof of concept to a full p5.js prototype.



2. Research and Design Goal Setting

A direct access to the target user group was not feasible, user research relied on secondary methods: literature review drawing on clinical sources such as Louis (2001), disability studies scholarship including Hendren (2020) and comparative analysis of

existing tools including Adobe Fresco and Procreate. A storyboard mapped the users journey speculatively, and co-design with adults with diagnosed motor conditions remains an essential next step.

People with motor differences are routinely excluded from mainstream creative practice. Widely used drawing tools correct movement toward an assumed standard of steadiness, positioning involuntary motion as a technical fault. Hendren (2020) challenges this framing directly, arguing that disability is a design problem rather than a personal limitation: the tools are wrong, not the users. This argument is foundational to MotionCanvas because rather than designing around tremor, the system is built around it.

The primary design framework guiding this project is Wobbrock et al. 's (2011) ability-based design, which explains systems should adapt to what users can do rather than requiring users to adapt to system defaults. MotionCanvas implements this through two specific mechanisms. First, an adaptive calibration phase observes each user's natural movement range during a 10 second live preview session, generating an individual movement profile that maps that range to the full canvas space. Second, a tremor reframe system applies a dual condition detection algorithm, high movement speed combined with low positional displacement, to identify tremor movement and translate them into crosshatch texture. MotionCanvas combines multiple input modalities - Leap Motion hand tracking, hover-and-hold dwell-based selection, mouse fallback, and keyboard shortcuts - with layered visual output channels; audio confirmation of gesture recognition was identified as a necessary output modality to reduce cognitive load but was not implemented.

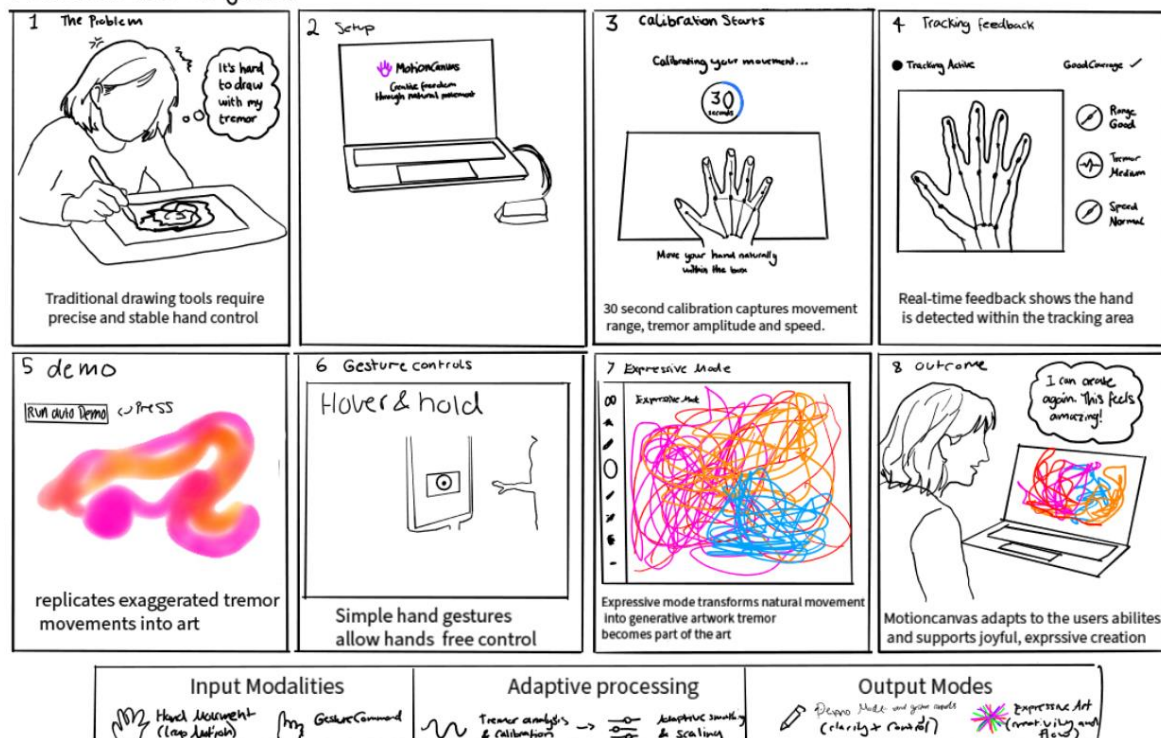
Dourish's (2001) embodied interaction theory provides further grounding: drawing is a bodily act and removing grip, and pressure requirements through gesture input preserves expressivity. Kendon (2004) supports this, arguing that gesture carries expressive meaning independent of precision. Hamraie (2017) and Barnes and Mercer (2003) extend this analysis toward the social model of disability, reinforcing that MotionCanvas should function as a creative platform rather than a rehabilitation device. Norman's (2013) affordance theory informed evaluation across all three iterations, particularly in identifying a non-functional tremor toggle in iteration 1 as a false affordance.

Three usability goals were established from this research:

1. Expressiveness – natural movement, including tremor, should produce visually engaging and intentional artwork
2. Adaptability – the system should personalise to each user's movement profile rather than applying fixed defaults

- Emotional engagement – users should experience the tool as a creative platform, not an accessibility device

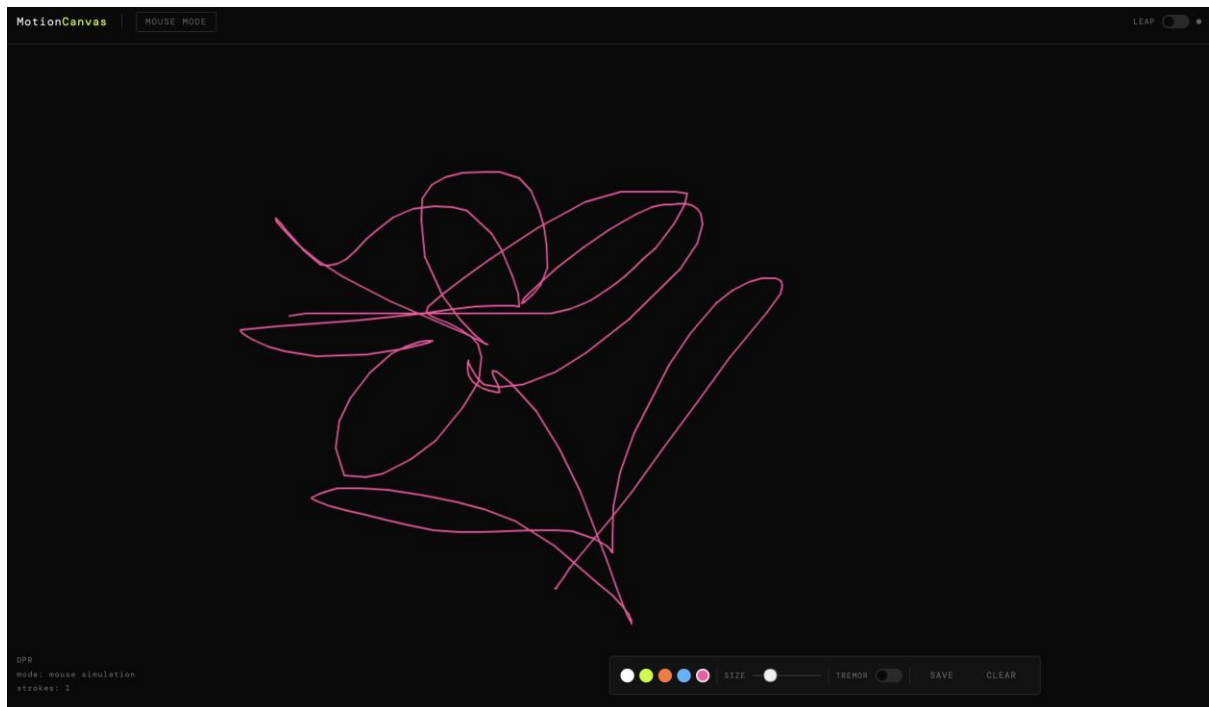
MotionCanvas - Storyboard



3. Design Process Documentation

Iteration 1 - Basic Drawing Mechanics

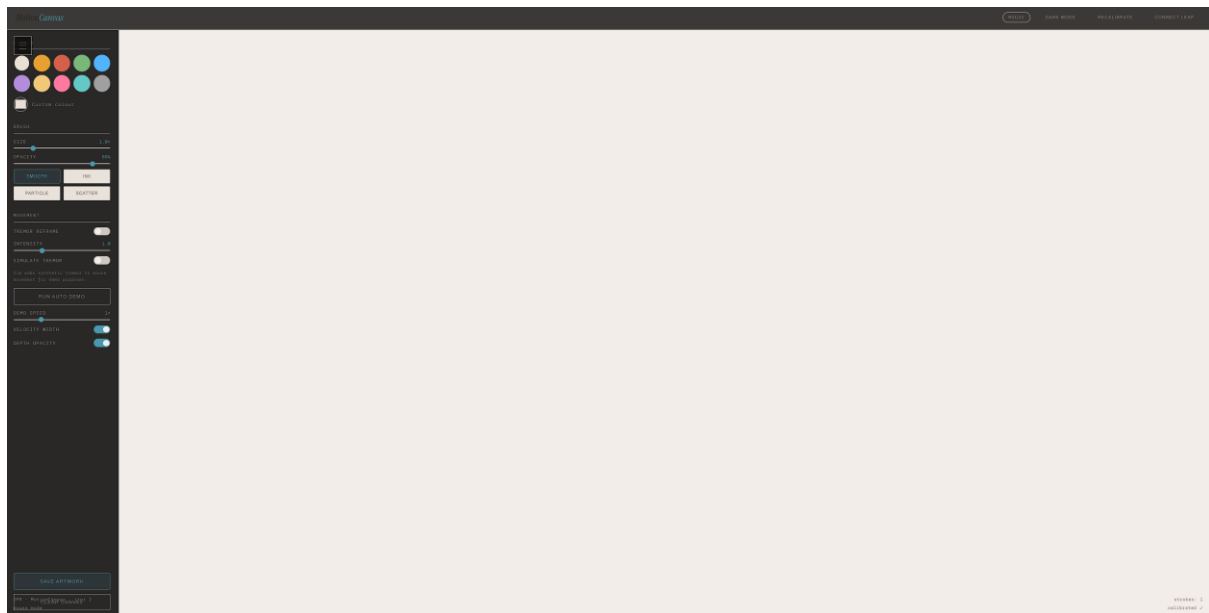
The design rationale for Iteration 1 was to verify the core technical concepts worked before adding complexity. The interface consisted of a HTML5 Canvas with JavaScript, mouse input, a line width slider, 5 colour swatches, Save and Clear buttons and a Leap Motion toggle. Evaluation was conducted as an internal group review. Core drawing mechanics functioned correctly; however, the tremor toggle was identified as a false affordance as it appeared interactive but produced no change and the central design concept could not be tested. The visual design was too generic and there was no onboarding or calibration stage. These findings led to Iteration 2: a full p5.js rebuild with functional tremor detection, a calibration stage and a structured onboarding screen.



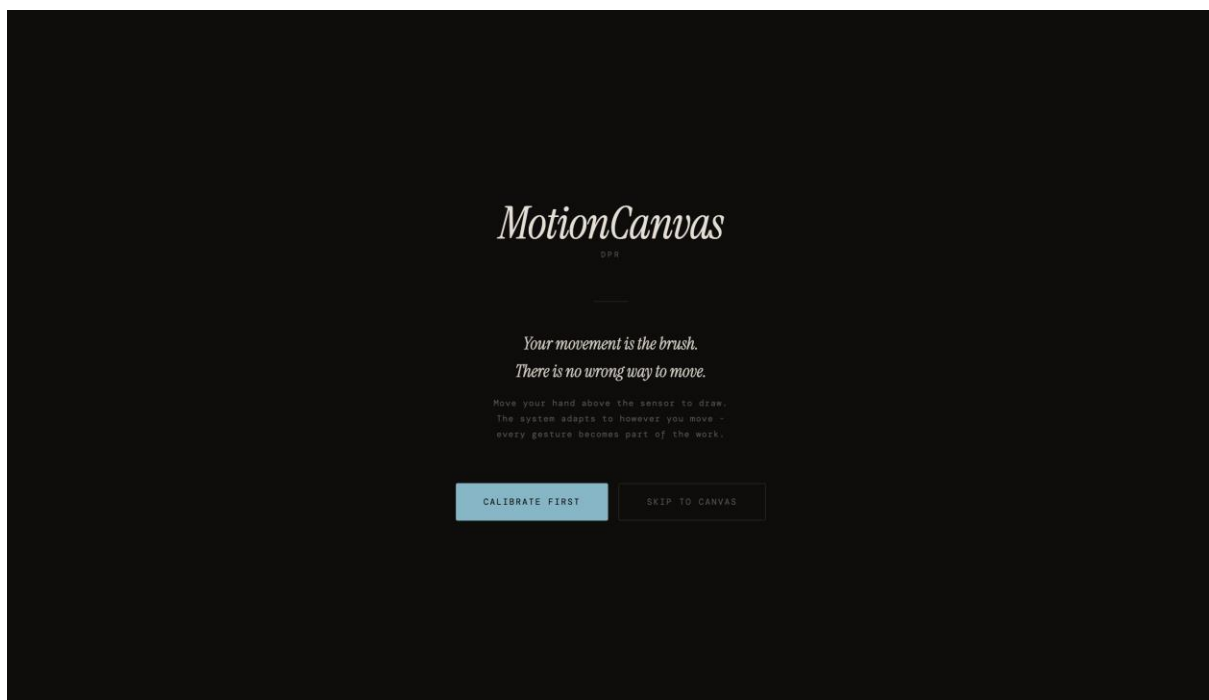
Iteration 2 - Prototype with Calibration and Tremor Detection

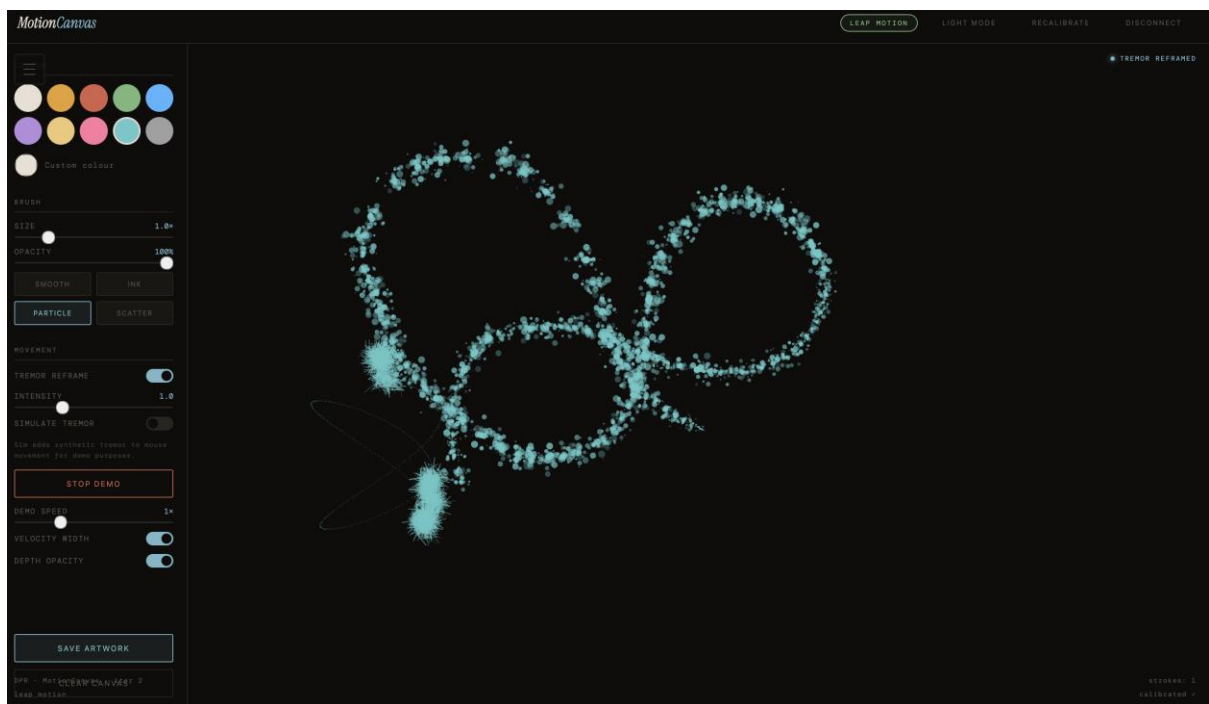
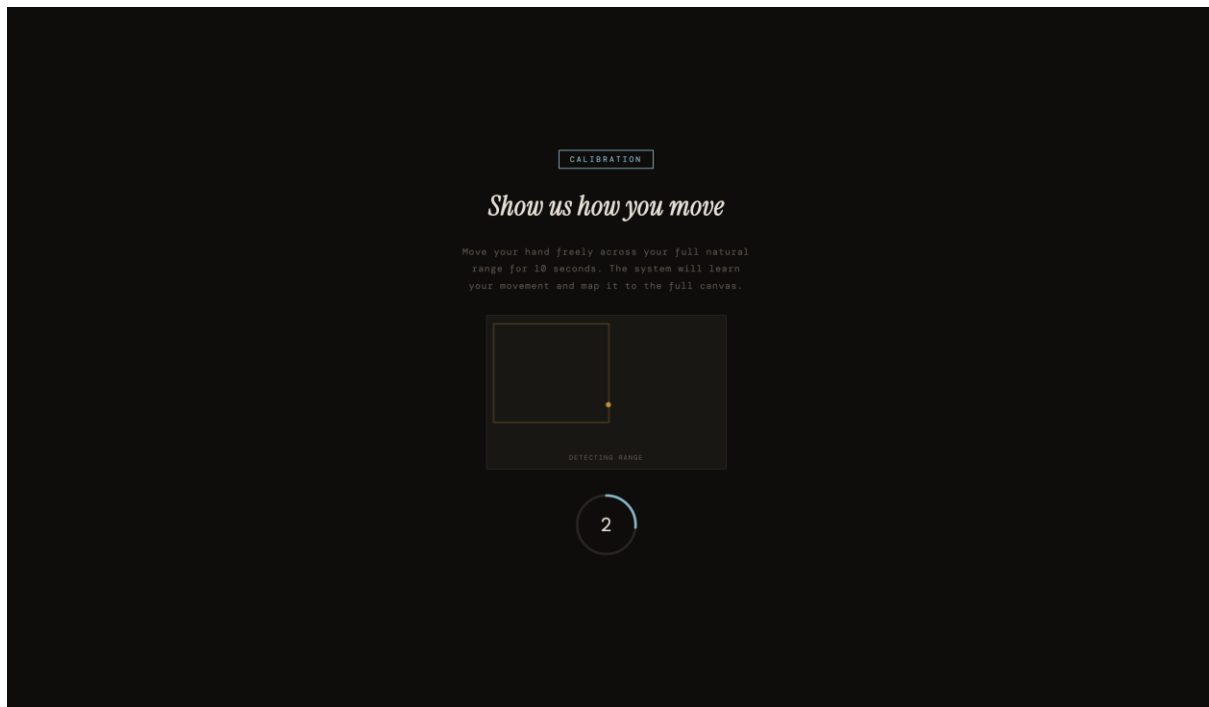
Iteration 2 was a complete rebuild in p5.js with a navigable screen flow:

Onboarding -> Calibration -> Canvas. The rationale was to make the tremor reframe concept functional and evaluable. The interface introduced four brush modes (Smooth, Particle, Scatter, Ink) a colour grid with custom colour picker, opacity and size sliders, functional tremor detection using a dual-condition algorithm (high speed, low displacement), crosshatch tremor reframing, a Stimulate Tremor toggle, auto demo mode and a sidebar for all brush, colour and movement controls. A light/dark mode toggle was partially implemented but the sidebar and top bar remained dark when the canvas switched to light.



A cognitive walkthrough identified four key issues: no undo function, calibration ended with no success state, sidebar lacked clear section grouping and custom colours could not save.

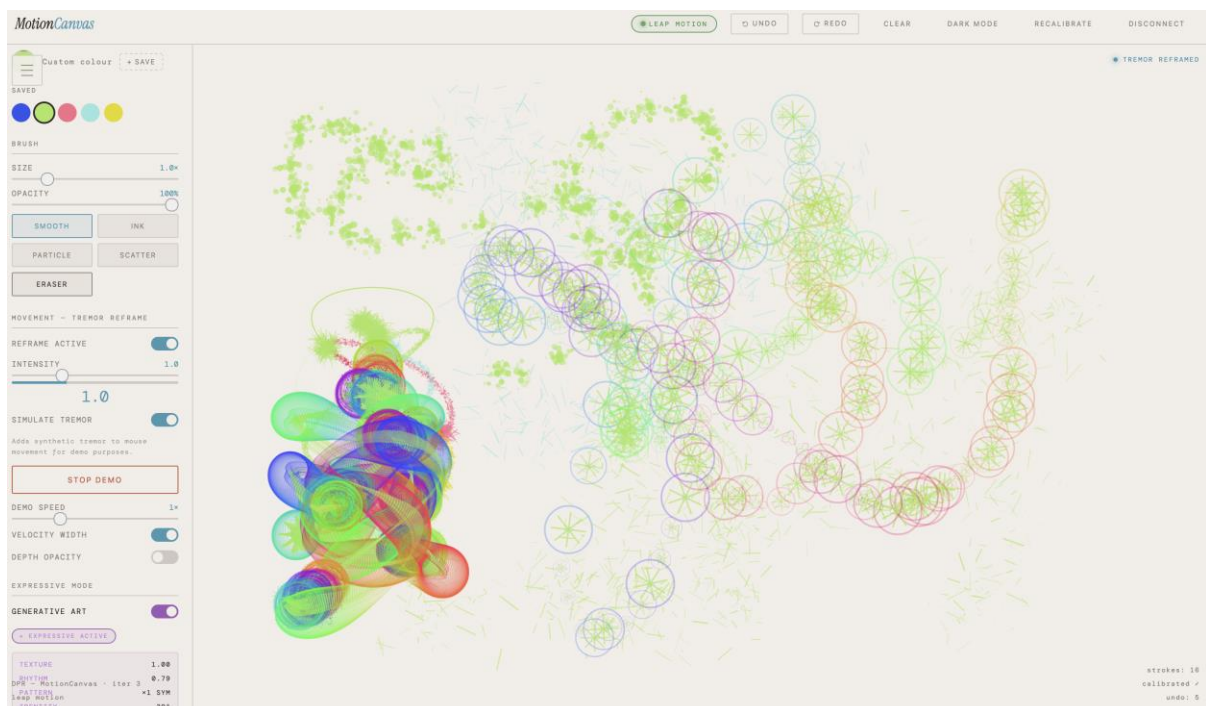
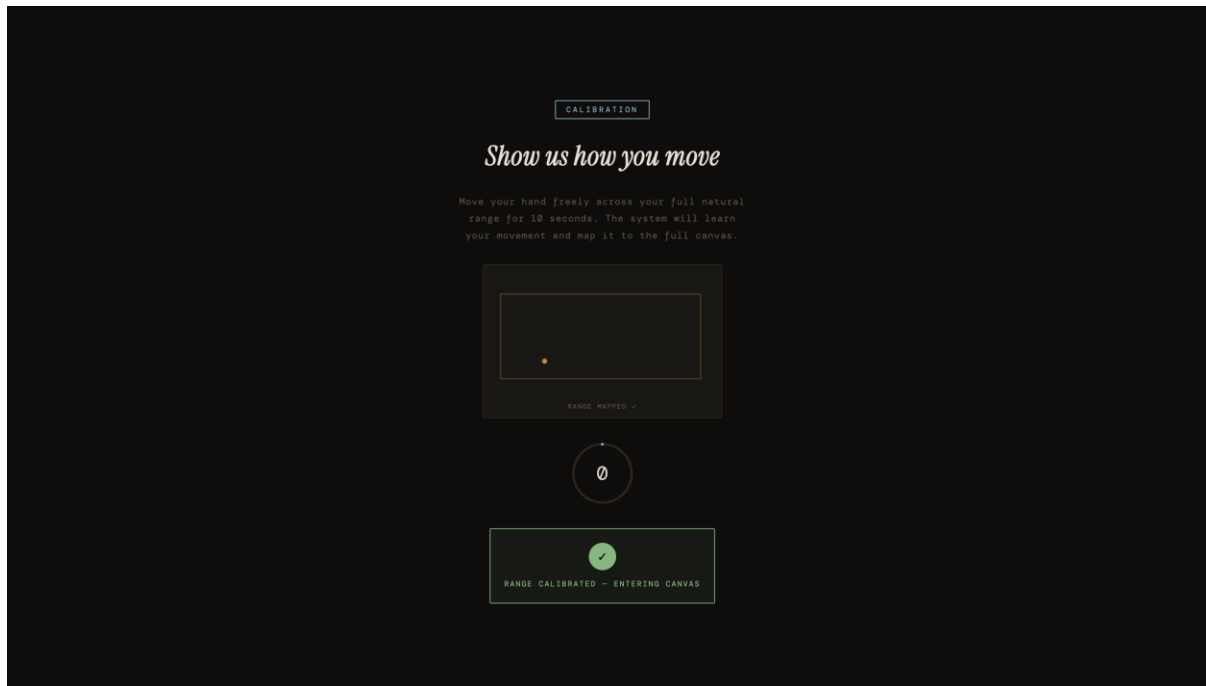




Iteration 3 - Usability Refinement and Submission Prototype

Iteration 3 focused on refining the prototype and solving the issues found in Iteration 2. Improvements included multi-step undo/redo, an eraser tool, a Clear button, an expanded colour palette and a consistent light mode across the interface. The Movement panel was renamed “Movement - Tremor Reframe” and enhanced with an intensity readout to better communicate the project’s core concept. An Expressive Mode was also added, transforming movement into abstract generative artwork. Evaluation showed that undo/redo significantly improved user control and error

recovery, while the revised panel made the tremor reframing concept more visible. . Hover and hold-based controls were used via leap motion sensor settings as an input method. But reliable testing with the physical sensor was not fully achieved due to hardware constraints (worked on some laptops not others). Audio feedback – short tones confirming hover and hold was identified as a necessary next step to add an auditory output channel alongside the existing visual feedback, which would create a genuine multimodal system.



4. Reflective Analysis and Inclusive Design Considerations

Affordances and Constraints

MotionCanvas responds directly to all three usability goals across its final iteration. Expressiveness is achieved through four distinct brush modes, crosshatch tremor reframing, particle-based rendering and an Expressive Mode that generates abstract generative art. Adaptability is realised through the calibration system, which personalises sensitivity and canvas mapping to each individual user's movement range rather than applying fixed defaults. The dual-condition tremor detection algorithm allows the system to distinguish between intentional fast movement and tremor without relying on a single metric. Emotional engagement is supported by the consistent visual identity, the warm palette, and the explicit labelling of the tremor reframe as a creative feature rather than an accessibility accommodation.

Several limitations emerged during development. The most significant is that Iterations 2 and 3 evaluations were conducted using the Simulate Tremor toggle, which generates tremor programmatically rather than capturing genuine involuntary movement. This does not accurately represent the rhythmic movement of real essential tremor, nor the unpredictable movement patterns of cerebral palsy or multiple sclerosis. The tremor detection algorithm has therefore not been validated against real user movement profiles. Co-design sessions with adults with diagnosed motor conditions would be essential before any further development.

An unexpected possibility emerged through the calibration system. Because each user's movement profile generates a unique mapping, two people with different tremor patterns produce entirely different visual outputs from the same interaction. MotionCanvas creates a movement signature, and the artwork is also a visual record of the user's individual motor identity. This was unplanned but represents a conceptual strength future iterations could develop.

Inclusive Design Considerations

MotionCanvas is designed for a range of motor differences, but different conditions present different interaction challenges. Users with mild essential tremor will likely find the Expressive brush modes immediately accessible, as small rapid movements generate dense textured marks. Users with more limited range of motion benefit most from the calibration system, which maps minimal movement to the full canvas. Users with multiple sclerosis may experience fatigue during longer sessions, removing grip strength through gesture input reduces physical effort, though future iterations could introduce optional session break prompts. Users with cerebral palsy may find the Particle and Scatter brush modes particularly suited to producing expressive output from unpredictable movement.

Cultural considerations are relevant. Abstract expressive art is not universally valued, and some users may not experience the tremor reframe output as meaningful creative work. The four brush modes offer some variation in output style, and future iterations could include modes with more representational output. Onboarding instructions are currently available in English only, which limits accessibility for non-English speaking users; translated onboarding would significantly broaden reach. The interface uses colour to communicate calibration state and mode selection so users with colour vision deficiencies may find some feedback harder to read, and additional non-colour indicators would improve accessibility.

Barriers to engagement include the hardware setup, which may require assistance for users with restricted movement. The gesture control vocabulary introduces a learning curve, and the panel reference list assumes active recall during drawing. Audio cues confirming hover-and-hold selection, would reduce cognitive load. Mouse mode, included as a full fallback throughout all three iterations, significantly broadens accessibility for users without sensor access.

MotionCanvas has the potential to promote creative access to people who have encountered consistent exclusion from art-making tools. Clift and Camic (2016) provide evidence for creative practice as a meaningful health and wellbeing intervention. The system is designed as a genuine creative tool, one that the user does not need to adapt to, because it has already adapted to them.

References:

- Barnes, C. and Mercer, G. (2003) *Disability*. Cambridge: Polity Press.
- Clift, S. and Camic, P. M. (eds.) (2016) *Oxford Textbook of Creative Arts, Health, and Wellbeing*. Oxford: Oxford University Press.
- Dourish, P. (2001) *Where the Action Is: The Foundations of Embodied Interaction*. Cambridge: MIT Press.
- Hamraie, A. (2017) *Building Access: Universal Design and the Politics of Disability*. Minneapolis: University of Minnesota Press.
- Hendren, S. (2020) *What Can a Body Do? How We Meet the Built World*. New York: Riverhead Books.
- Hurst, A. and Tobias, J. (2011) 'Empowering individuals with do-it-yourself assistive technology', *Proceedings of the 13th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '11)*, pp. 11–18. New York: ACM.
- IDEO (2019) *Painting with Code*. IDEO Journal.

- Kendon, A. (2004) *Gesture: Visible Action as Utterance*. Cambridge: Cambridge University Press.
- Louis, E. D. (2001) 'Essential tremor', *The Lancet Neurology*, 1(7), pp. 12–20.
- Microsoft Design (2016) *Inclusive Design Toolkit*. Microsoft Corporation.
- Nielsen, J. (1994) *Usability Engineering*. San Francisco: Morgan Kaufmann.
- Norman, D. (2013) *The Design of Everyday Things*. Revised edn. New York: Basic Books.
- Siebers, T. (2010) *Disability Aesthetics*. Ann Arbor: University of Michigan Press.
- Wobbrock, J. O., Kane, S. K., Gajos, K. Z., Harada, S. and Froehlich, J. (2011) 'Ability-based design: concept, principles and examples', *ACM Transactions on Accessible Computing*, 3(3), pp. 1–27.